CAMPUSX · 100 DAYS OF MACHINE LEARNING

Machine Learning Notes

Feature Engineering and Missing Data

Feature engineering and missing-data notes · Chapters 31–40

Rebuilt as a book: mechanism first, prose before lists, every example traced

Study notes from the CampusX 100 Days of Machine Learning playlist. Every claim is tied to the lecture, Scikit-learn behaviour, or a worked calculation.

Contents

Power Transformer 3
31.1 The problem with a long tail 3
31.2 Box–Cox 4

CHAPTER 31

Power Transformer

Skewness, Box-Cox and Yeo--Johnson

WHAT THIS CHAPTER BUILDS

This chapter starts from the failure mode: a long tail. It then derives the transform, shows what lambda does, and ends with a traced Scikit-learn pipeline that must be fitted on training data only.

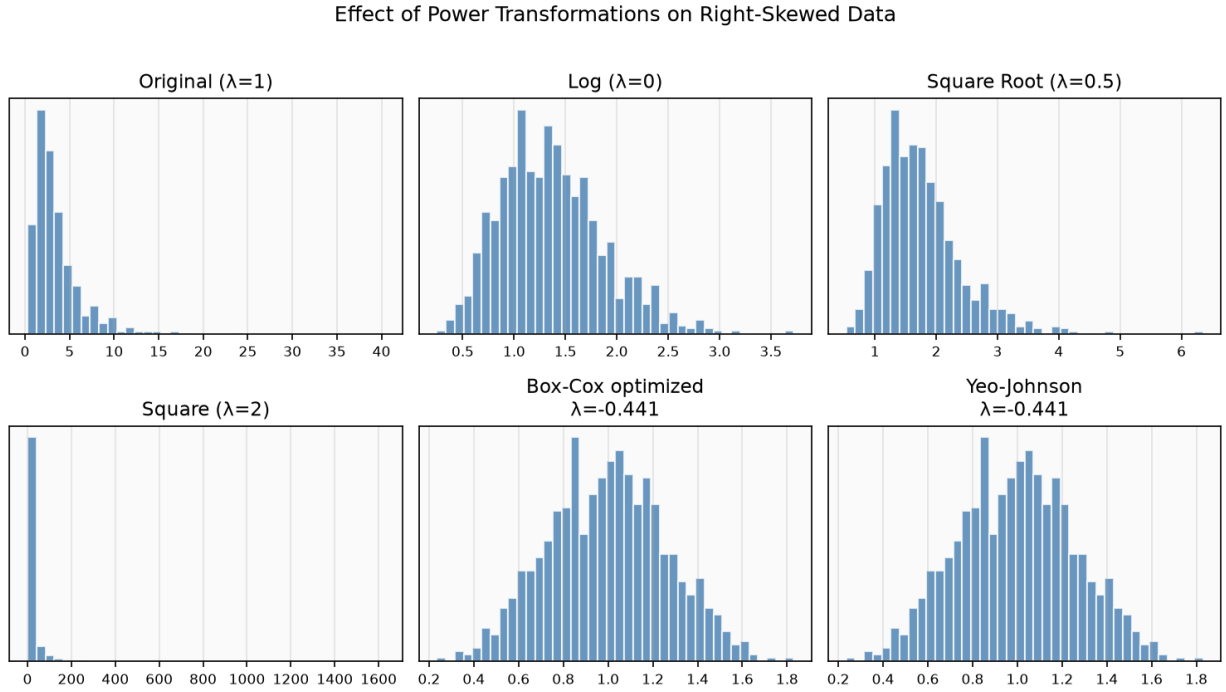
31.1 The problem with a long tail

A feature with a long right tail is not merely ugly. It changes which observations dominate a squared-error loss, it stretches distances, and it makes a mean a poor summary of the typical case. Consider ten incomes:

Value	Deviation from mean
10	−25.4
11	−24.4
12	−23.4
90	+54.6

The mean is not robust

One large value moves the mean, and squared error moves it even more. The median stays near the bulk. A transform does not make the outlier disappear; it changes how much leverage the outlier has.



Power transforms change the shape of the distribution while preserving order. The same data is shown before and after several values of lambda.

31.2 Box–Cox

Box–Cox transform

For positive x,

$$y^{(\lambda)} = \begin{cases} \frac{x^{\lambda}-1}{\lambda} & \text{if } \lambda \neq 0 \\ \ln x & \text{if } \lambda = 0 \end{cases}$$

The case at zero is the continuous limit, not an arbitrary exception.

```
from sklearn.preprocessing import PowerTransformer
pt = PowerTransformer(method="box-cox")
X_train_t = pt.fit_transform(X_train[["income"]])
X_test_t = pt.transform(X_test[["income"]])
```